

Hyperelliptic cover infrastructure: theorem, implementation, and transfer audit

aburan28/crypto research record

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Hyperelliptic cover infrastructure audit

Outcome

The repository has replayable **curve-level** hyperelliptic cover constructions for two catalog model families, but the audited evidence does not establish a general cover-to-Jacobian transfer pipeline or a full Jacobian index-calculus solver.

The precise correction is:

- Xavier Xarles’s Corollary 3 is a theorem about a hyperelliptic curve with **two independent maps** to a given elliptic curve over the same field. It is not the trivial observation that an elliptic curve has genus one.
- Crypto PR [#1383](#) gives stronger, simpler one-map constructions for the two catalog families and verifies their geometry. Its original head contains 115 certificates. The catalog at the audit baseline contains 121 because six models were added later.
- Crypto PR [#1389](#) and cryptanalysis PR [#290](#) implement a bounded experiment on a genus-three **non-hyperelliptic plane quartic**. Its linear system is formed after elliptic norm projection. It is not full arithmetic or index calculus in the quartic Jacobian.
- The odd-characteristic Mumford implementation audited at commit [5874ce8b393dccc4c735cc7ca5a47de7fe196c59](#) is for odd-degree, one-rational-infinity models. The prime constructor in PR [#1383](#) emits a monic genus-two sextic with two rational points at infinity. Applying that implementation to this even-degree model is unsupported.
- The delivered implementation at commit [3b8664a466efd56121509cd77866a07e3cb55741](#) adds checked odd-prime and binary one-infinity arithmetic and a **bounded** binary transfer: rational elliptic point-class pullback, rational source-point-divisor pushforward, and typed [3] composition witnesses, including infinity and the exceptional ramified fiber. It explicitly rejects an arbitrary $\text{Jac}(H)$ pushforward.
- No audited record supplies an arbitrary-class pullback/pushforward pair, an exact image-generator replay, and a subgroup log-recovery certificate. Cover existence and the bounded transfer therefore do not establish a full discrete-log transport pipeline.

All transfer assessments below use exactly **unknown**, **supported**, **refuted**, or **not_applicable**. Missing evidence is **unknown**, not a nonexistence result.

Audit basis

The repository audit is pinned to the following immutable revisions and content digests.

- **Crypto audit baseline** — existing arithmetic and current generated catalog:
[5874ce8b393dccc4c735cc7ca5a47de7fe196c59](#)
- **Hyperelliptic infrastructure implementation** — checked arithmetic, bounded binary transfer, standalone command, and release integration:
[3b8664a466efd56121509cd77866a07e3cb55741](#)

- **PR #1383** — original cover certificates:
head 7c186ca89b8efeb0db2cea53e10e6da520166b56
merge c8a8f027998295f2a3620d96a2f0985f17ffb52b
- **PR #1389** — crypto quartic experiment:
head 6c6247bde5e585642773ed5b6b933b4ec4c6fd4f
merge 7ea1db7a6b4b6962f3fa38d70ac65300d25ceb4b
- **PR #290** — cryptanalysis mirror; PR base was claude/cryptanalysis-repo-setup-p59kek:
head 3b14ec30de6c79fbc9dd72f4ed3eb8ea19d5a642
merge 54cbf1dc27879b9bd441df7b64e2c629e9d9d9ea
- **Xarles source** — primary theorem source:
arXiv:1303.4220v1, 2013-03-18
- **Xarles v1 PDF SHA-256** — retrieved 2026-10-06:
5bf067b8e6376eca4559bc7335d824c3f74911649bd806bfcda1cefb0d6ce2ed
- **Xarles v1 source-gzip SHA-256** — retrieved 2026-10-06:
243c91375bd9bae5c8e17093fcd8e9e4af12b362beaf8a1cd0e4f7033206e51
- **PR #1383 covers.json SHA-256** — exact 115-row artifact:
5f90fe0c1dfaa5cd2bd187a87dfd5ee5985c4d0171ea93e7874f9246527aaf9b
- **Current covers.json SHA-256** — exact 121-row artifact at the audit baseline:
b1634d75fff27add2e27adf996ba820a8852ea72468088d9eb3c88bb69759a39
- **Quartic kernel SHA-256** — byte-identical at both PR heads:
54ef65c7325a7c850741a8def84ee2b2ceddccc3689a0879afd2796eb7cb01e52
- **Quartic CLI SHA-256** — byte-identical at both PR heads:
0f71936014b3ff54dacc9dc6b642c88bea60b70a658f6e42c467aab73f11a069
- **Quartic receipt SHA-256** — byte-identical at both PR heads:
d3de82409e6fe82a59e74425916ed820ac5f8912e74046f41468cef56c01a967

The receipt retained in PR #1389 says `repository-CI-pending`; the later PR description reports the final head passed its applicable CI. These are different historical statements. The receipt has not been rewritten to conceal that ordering.

Requirement-to-evidence table

Requested item	Status in this report	Evidence or exact gap
Primary theorem and exact hypotheses	Verified complete	Xarles v1, Theorem 2 and Corollary 3, audited below
Audit PRs #1389 and #290	Verified complete	Exact heads, matching source hashes, frozen correctness receipt
Audit PR #1383 certificates	Verified complete	Exact head artifact: 28 prime plus 87 binary records
Separate odd-characteristic and characteristic-two models	Specified	Infrastructure guide and additive JSON Schema
Constructive cover algorithms	Partial in repository	Two PR #1383 families are supported; this is not a theorem-complete implementation
Genuine Jacobian arithmetic	Partial	The committed feature checks odd-prime and binary one-infinity models; prime even-sextic cover arithmetic remains structured unsupported
Pullback and pushforward	Partial	Bounded binary rational-point domains are supported; arbitrary Jacobian pushforward and the prime balanced model are not implemented

Requested item	Status in this report	Evidence or exact gap
Subgroup preservation	Partial	The exact F_2^7 order-29 catalog subgroup has a bounded certificate; other catalog subgroups and arbitrary Jacobian transfer remain unknown
Catalog schema	Implemented in this change	<code>docs/curves/hyperelliptic-infrastructure.s</code>
Standalone command contract	Implemented and documented	Direct commands with strict model shapes, 32 MiB input caps, propagated output errors, required external registry for catalog export, and no <code>cargo run</code> requirement for packaged builds
Full Jacobian IC solver	Not attempted; separate deliverable	Acceptance gates are specified below

Xarles Theorem 2 and Corollary 3

Primary source: Xavier Xarles, *Hyperelliptic curves covering an elliptic curve twice*, arXiv:1303.4220v1 ([abstract](#), [PDF](#)).

Exact statement and hypotheses

Theorem 2 assumes:

- an arbitrary field K with $\text{char } K \neq 2, 3$;
- $j \in K$ with $j \neq 0, 1728$; and
- $A = 3^3 j / (2^2(j - 1728))$.

It takes

$$E_A : y^2 = x^3 - Ax + A$$

and gives the genus-five hyperelliptic curve

$$H_A : y^2 = A(x+1)^4(x^2+1)^4 - 2^6 x^3(x^2+x+1)^3$$

with two independent maps to E_A . In the proof, E_A is replaced by the isomorphic genus-one quartic $D : y^2 = x^4 + x^3 + B$, where $B = A/2^6$. The auxiliary curve C has maps $f_1(x, y, z) = (x, y)$ and $f_2(x, y, z) = (z, y)$ to D . The proof states that both maps have degree three.

Corollary 3 states: for every elliptic curve E/K , where $\text{char } K \neq 2, 3$, there is a hyperelliptic curve H/K of genus at most five and two independent maps $H \rightarrow E$, all defined over K . For $j \neq 0, 1728$, the proof uses the same quadratic twist of the Theorem 2 construction. For $j = 0$ or 1728 , it cites known genus-two constructions rather than writing their equations in this note.

What follows and what does not

Question	Audit result
Base field	Any field K , subject to characteristic not 2 or 3
Field extension required	No; the corollary says H and both maps are defined over K

Question	Audit result
Genus	Exactly 5 in Theorem 2; at most 5 in Corollary 3, with genus 2 cited for exceptional j
Number of maps	Two independent maps, not merely one cover
Degree	The proof gives degree 3 for both maps in the nonexceptional case
Separability	Not a separately stated hypothesis or conclusion. In the nonexceptional case, degree 3 and characteristic not 3 imply the degree-three maps are separable. The note does not separately audit the exceptional- j maps.
Constructive status	Formula-level constructive for $j \neq 0, 1728$, after a quadratic twist and standard isomorphisms. It is not a uniform executable algorithm: final coordinate formulas on the displayed H are not fully expanded, and the exceptional cases are cited rather than constructed.
Smoothness certificates	Asserted in the theorem through the smooth hyperelliptic curve; no machine-checkable certificate format is supplied
Minimal genus	Not established. The paper says genus 2 is unavailable for a general elliptic curve over a general field for the two-map problem.

The word “hyperelliptic” here is substantive. The displayed H_A is a genus-five curve with a degree-two map to a line. The identity map $E \rightarrow E$ is neither the construction nor two independent maps, and under the standard convention a hyperelliptic curve has genus at least two.

The result is also stronger than the one-map existence question. For odd characteristic short Weierstrass models, PR #1383 uses the elementary quadratic pullback $x = u^2 + c$ to obtain a genus-two cover with one map. That does not replace Xarles’s two-independent-map statement.

Repository implementation audit

PR #1383: explicit hyperelliptic covers

At the exact PR head, `docs/curves/covers.json` contains 115 rows:

Family	Count	Source curve and map	Verified scope
Prime short Weierstrass, characteristic > 3	28	$H : v^2 = F(u^2 + c),$ $(u, v) \mapsto (u^2 + c, v)$	Genus 2, degree 2, separable, same field
Ordinary binary	87	$H : v^2 + u^2v =$ $u^7 + au^4 + du,$ $(u, v) \mapsto (u^3, uv + d),$ $d^2 = b$	Genus 3, degree 3, separable, same field

The verifier checks canonical field/model inputs, the defining-equation identity, the family-specific smoothness and genus argument, map degree, and behavior at infinity. Tests enumerate all affine cover points for all nonsingular short Weierstrass models over $\mathbf{F}_5, \mathbf{F}_7, \mathbf{F}_{11}$ and all ordinary models over $\mathbf{F}_8$, and reject altered certificates and identity bindings.

Important boundaries:

- the fixed 20-base Miller–Rabin screen is not a primality proof; prime field status remains a registry assumption;
- the binary modulus check uses the exact Rabin irreducibility criterion;
- minimum genus, minimum degree, subgroup transport, and DLP advantage are not established;
- a construction failure is recorded as unsupported or invalid input, never as universal nonexistence.

The current main artifact has 121 verified rows, not 115. Its catalog `checker_source_sha256` digest (checker module UTF-8 followed by the `curve_cover_check` wrapper UTF-8) remains 205e2b9b250568c2b3235c2fd803122a085505798c28fee7e93ce08e0b6d3cc0; the registry expanded after PR #1383. This report preserves both populations and does not rewrite the original PR count.

PRs #1389 and #290: bounded plane-quartic norm projection

These PRs use

$$C : v^4 = x^3 + ax + b, \quad \pi(x, v) = (x, v^2)$$

over small prime fields. A smooth plane quartic has genus three and is non-hyperelliptic. The implementation certifies complete line sections on C , then projects their divisor relations to the elliptic factor.

The frozen fixture reports 45 rational quartic points, 79 certified sections, 22 distinct nonidentity norm images, 11 folded matrix columns, and rank 11. Its retained controls cover 118 known-answer scalar cases and all 990 secant pairs for the fixture. Those are correctness counts, not a common operation unit or a performance measurement.

The matrix columns identify points only after elliptic norm projection. Deck-conjugate divisors can have equal norm and different Prym components. Accordingly:

- the claim “the PRs implement the bounded elliptic norm-projection experiment” is **supported**;
- the claim “the PRs construct a hyperelliptic cover” is **refuted**;
- the claim “their matrix is a factor-base relation matrix for the full quartic Jacobian” is **refuted**; and
- full-Jacobian target decomposition, a reduced factor base with large-prime recombination, extension-field descent, and end-to-end performance remain **unknown** or unimplemented.

Passing the PR tests proves the bounded contracts those tests exercise. It does not promote the experiment to a general Jacobian solver.

Existing Jacobian arithmetic at the audit baseline

The odd-characteristic module contains actual Cantor/Mumford operations: validity checking, reduction, equality, addition, negation, and scalar multiplication. Its sound documented model is monic squarefree $\deg f = 2g + 1$, with one rational point at infinity. At the pinned baseline:

- the constructor did not fully validate primality, squarefreeness, degree, genus, or domain compatibility;
- point embedding rejected $y = 0$, even though the corresponding Weierstrass/2-torsion divisor is representable;
- targeted nontrivial shared-support, exceptional-divisor, closure, associativity, malformed-input, and independent operation-oracle tests were missing; and
- one genus-two order check compared point-count/L-polynomial arithmetic with exhaustive Mumford enumeration, but this is not an independent addition oracle.

Most importantly, the prime PR #1383 cover is a monic sextic. Its two rational points at infinity require balanced or real-model Jacobian arithmetic. The existing one-infinity representation is a different mathematical object and must return structured unsupported for this input.

The characteristic-two module represents $v^2 + h(u)v = f(u)$ with a one-infinity Mumford model. At the audit baseline, its public constructor did not globally prove smoothness and its direct unit tests were concentrated

on a genus-one generic-point fixture. Those are baseline findings; the separately identified committed feature below does not rewrite them.

The specialized `jv_cover` module contains a conorm–norm transfer for its own Joux–Vitse genus-three family. It is not a generic implementation of the PR #1383 morphisms, is not a replacement for record-specific pullback/pushforward certificates, and does not establish support for the prime even-sextic family.

Delivered implementation evidence after the audit baseline

The following evidence is bound to implementation commit `3b8664a466efd56121509cd77866a07e3cb55741`, whose source history is based on the audit baseline. It does not rewrite what was true at the baseline and the test results are local receipts, not CI receipts. Root independently ran the listed targets on 2026-10-06 with the Rust 1.90 workspace toolchain.

Capability	Exact supported domain	Feature evidence	Assessment
Odd-prime Jacobian arithmetic	Deterministically validated odd prime $p < 2^{64}$, monic squarefree $y^2 = f(x)$, $\deg f = 2g + 1$, one rational infinity	Checked field/curve-bound Mumford constructors and exact Cantor divisions; all 24 classes of $y^2 = x^5 + x^2 + 1$ over $\mathbf{F}_3$, including two-torsion and nontrivial shared support; order 24 independently derived from $\mathbf{F}_3/\mathbf{F}_9$ point counts	supported in this domain; the PR #1383 even sextic is refuted as an input to this implementation
Prime reference counters	Linear enumeration for $p \leq 1,000,000$; quadratic enumeration and checked $\mathbf{F}_{p^2}$ context for $p \leq 4096$	Deterministic primality, validated nonzero quadratic nonresidue, canonicalized extension elements, and explicit enumeration bounds	supported only as bounded diagnostics; not a large-field point counter
Binary Jacobian arithmetic	Polynomial-basis $\mathbf{F}_{2^m}$, $1 \leq m \leq 4096$, Rabin-verified modulus, separable $v^2 + h(u)v = f(u)$, $\deg f = 2g + 1$, $\deg h \leq g$, one rational infinity	Checked canonical operations; noncanonical trailing-zero polynomials rejected; exhaustive 486-class F8 genus-three fixture; retained and cancelling shared-support branches	supported in this domain; even-degree/generalized infinity models are outside scope
Binary pullback	$[P] - [O]$ for rational $P \in E(K)$ in <code>binary_cubic_pullback_vt</code>	Typed generic, infinity, and exceptional two-torsion fibers with reduced Mumford output	supported for the declared point-class domain
Binary pushforward	$[Q] - [\infty_H]$ for rational $Q \in H(K)$, plus the implementation's own three typed pullback witnesses	Direct point map and ramification-aware formal fiber sum	supported only for this bounded domain

Capability	Exact supported domain	Feature evidence	Assessment
Binary composition	The preceding rational elliptic point classes	$\pi_*\pi^* = [3]$, exhaustively checked on rational target points for all 56 ordinary F8 models, including infinity and the exceptional ramified point	supported for the declared point-class domain
Exact binary subgroup	EC1N7Ce0hb6d297a2ca08, generator $(0x26, 0x5)$, order 29	Independently checks $G \neq O$ and $[29]G = O$, so primality of 29 proves the supplied integer is the exact order; then checks nonidentity pullback image, $[29]\pi^*G = 0$, degree inverse $3^{-1} = 10 \bmod 29$, and $[10](\pi_*\pi^*G) = G$	supported for this exact subgroup and bounded transfer domain; the generic order helper is conditional on independent exact-order evidence
Arbitrary binary Jacobian norm	General reduced Mumford class in $\text{Jac}(H)$	Public negative control returns <code>Unsupported(ArbitraryJacobianPushforward)</code>	implementation-support claim refuted
Standalone wrapper	Supported explicit model shapes; no implicit registry lookup	Unknown model fields rejected, all inputs capped at 32 MiB, stdout failures propagated, required external registry for catalog export, and combined wrapper/checker source provenance	supported as an interface contract; no new mathematical claim

The 64-bit prime bound belongs to ordinary prime-curve and group arithmetic. It does not enlarge the deliberately small reference counters: linear enumeration stops at one million, while both the quadratic counter and `Fp2Ctx` stop at 4096. The checked extension context revalidates the odd prime and its nonzero quadratic nonresidue; its operations reduce public element coordinates before use.

The standalone commands consume a supported model object directly. They do not resolve an ICV1 slug or read a registry unless `catalog-export` is explicitly invoked with its required `--registry` path. Catalog export names the wrapper in `generated_by` and hashes the checker source concatenated with the wrapper source in `producer_source_sha256`.

The content pins are listed one source or producer per digest so the complete 64-character values remain readable in the PDF:

- `src/prime_hyperelliptic/curve.rs`
7a769bfb7abfe92ce06da7c02340f2a2426827d8066ebef510375149b8c3d5ad
- `src/prime_hyperelliptic/fp2.rs`
b833bdff5247f2eea0cde7feb89d0378cfe8305671f9065d2dae7ab538c03e55
- `src/prime_hyperelliptic/fp_poly.rs`
b5d1a1ba5e95d8d0dd06b71757567fb559adbdd09b1ab623ae76e0049ee593c2
- `src/prime_hyperelliptic/mod.rs`
b6906f75b4d0eb6b7ac406a0fca1b2cef1786a528d4aedbabb414e9fa1e7f7662

- `src/binary_ecc/hyperelliptic.rs`
09ce8a689aa8e190f51c871f9e94467271ea731b9629f69d322ae716de507c5c
- `src/binary_ecc/cover_transfer.rs`
67b94112f70becf100ccc2cc7e0b3bc057e5663a4019b63fe966de9e62fd5069
- `src/binary_ecc/mod.rs`
84ece02cbf6f897317b236239c47efb0738d0e8e9f3d8530d2d4793609426099
- `src/bin/curve_cover_check/checker.rs`
65a51b7265d1a426b915deffc58c34562204c8f649219ce50deaed5cb8d23f18
- Existing catalog producer concatenation (checker, then `curve_cover_check` wrapper)
205e2b9b250568c2b3235c2fd803122a085505798c28fee7e93ce08e0b6d3cc0
- `src/bin/hyperelliptic_cover.rs`
cd32103c2ea0f23ca0dc568155e836cc055a854f23975a37314b63f5aa8361a1
- New catalog producer concatenation (checker, then `hyperelliptic-cover` wrapper)
3927efadd7e7e966bb284affc56ac22a823f98f498017509e266b603b244e37

Local command	Result	Test time
<code>cargo test --lib</code>	10 passed, 0 failed, 1 ignored	36.18 s
<code>prime_hyperelliptic::curve::tests</code>		
<code>cargo test --lib</code>	6 passed, 0 failed, 0 ignored	0.00 s
<code>prime_hyperelliptic::fp2::tests</code>		
<code>cargo test --lib</code>	11 passed, 0 failed	29.07 s
<code>binary_ecc::hyperelliptic::tests</code>		
<code>cargo test --lib</code>	4 passed, 0 failed	0.27 s
<code>binary_ecc::cover_transfer::tests</code>		
<code>cargo test --bin</code>	6 passed, 0 failed, 0 ignored	0.33 s
<code>curve_cover_check</code>		
<code>cargo test --bin</code>	6 passed, 0 failed	0.00 s
<code>hyperelliptic-cover</code>		

The F3 point-count cross-check independently certifies the group order, not each addition output. The F8 exhaustive checks are deterministic internal controls, not an external CAS oracle. These distinctions keep the feature evidence useful without overstating independence.

Typed correspondence graph

Editable source: [typed_correspondence.dot](#). Vector rendering: [typed_correspondence.svg](#).

The graph separates a curve morphism from the two induced Picard/Jacobian homomorphisms. Green Jacobian edges are labeled **supported-with-domain**: they cover only binary rational point classes, rational source-point divisors, and the transfer implementation’s typed fiber witnesses. The red arbitrary-class norm is not hidden behind those edges. The graph also keeps the non-hyperelliptic quartic experiment on a separate path.

Transfer obligation table

ID	Typed claim or obligation	Status	Evidence and remaining work
T1	Xarles Corollary 3 gives a same-field genus- ≤ 5 two-map existence theorem in characteristic not 2 or 3	supported	Primary source, Corollary 3

ID	Typed claim or obligation	Status	Evidence and remaining work
T2	Xarles gives a uniform executable construction for every input model	unknown	Nonexceptional formulas need full coordinate expansion and replay; exceptional j cases are cited
T3	PR #1383 prime and ordinary-binary records contain explicit same-field $H \rightarrow E$ maps	supported	Native checker, exact 115-row PR artifact
T4	PR #1383 verifies smoothness, genus, equation identity, degree, separability, and infinity behavior within its families	supported	Checker and exhaustive small-field controls
T5	A failed bounded constructor proves no cover exists	refuted	Both the checker contract and this audit prohibit that inference
T6	The current one-infinity odd-characteristic arithmetic supports PR #1383's even sextic	refuted	Two rational infinity points versus a one-infinity Mumford model
T7	The committed feature implements checked odd-degree/one-infinity arithmetic in its declared prime-field model	supported	Deterministic primality for every supported modulus below 2^{64} , typed constructors/errors, exact divisions, two-torsion/shared-support cases, and all 24 F3 classes; the L-polynomial point-count route independently gives group order 24, not an addition oracle
T7a	The prime reference counters are general large-field algorithms	refuted	Linear enumeration is bounded to $p \leq 1,000,000$; quadratic enumeration and checked Fp2Ctx are bounded to $p \leq 4096$
T7b	The same odd-prime implementation has independent operation-oracle certification across every advertised genus and field size	unknown	The bounded F3 evidence does not justify that broader claim

ID	Typed claim or obligation	Status	Evidence and remaining work
T8	Characteristic-two genus-three cover arithmetic is checked for the PR #1383 binary family	supported	Committed feature; canonical/trailing-zero rejection, 486-class F8 exhaustive fixture, and explicit shared-support branches
T9	Pullback of rational elliptic point classes for <code>binary_cubic_pullback_v1</code> , including infinity and exceptional two-torsion	supported	Checked reduced Mumford construction and exhaustive F8 rational-point controls
T9b	Pullback for the prime two-infinity family or other unlisted cover families	unknown	No balanced prime implementation or certificate
T10	Pushforward of rational binary source-point divisors and the transfer implementation's typed pullback witnesses	supported	Direct point map and bounded ramification-aware formal fiber sum
T10b	A verified pushforward/norm for an arbitrary binary $\text{Jac}(H)$ class is implemented	refuted	The public negative-control API returns <code>Unsupported(ArbitraryJacobianPushf</code>
T11	$\pi_*\pi^* = [3]$ is replayed on the supported binary rational point-class domain, including infinity and the exceptional divisor	supported	<code>Unsupported(ArbitraryJacobianPushf</code> Exhaustive rational target-point replay across all 56 ordinary F8 models
T11b	The feature supplies a pushforward on arbitrary $\text{Jac}(H)$ classes	refuted	The supported composition verifier is intentionally limited to its typed point-fiber witnesses
T12	The exact order-29 subgroup in <code>EC1N7Ce0hb6d297a2ca08</code> meets $\ker \pi^*$ trivially and supports inverse-degree recovery	supported	Independently verify $G \neq O$ and $[29]G = O$, hence exact order 29; then verify the nonidentity order-29 image, $3^{-1} = 10 \bmod 29$, and $[10](\pi_*\pi^*G) = G$. The generic helper alone is only conditional on a separately verified exact order.
T12b	Every PR #1383 subgroup has a corresponding kernel/image/recovery certificate	unknown	Only the named F2^7 order-29 record has the delivered bounded certificate

ID	Typed claim or obligation	Status	Evidence and remaining work
T13	A nontrivial field extension is needed by the PR #1383 constructors	not_applicable	Both constructors are over the declared target field; record the degree-one identity relationship
T14	PRs #1389/#290 establish full-Jacobian relations	refuted	Relations are used only after elliptic norm projection
T15	Any measured computational advantage follows from cover existence	unknown	No end-to-end matched measurement; all performance fields remain null
T16	The standalone wrapper performs implicit registry lookup or accepts arbitrary model fields	refuted	Supported model shapes are explicit; unknown fields are rejected, and only catalog export reads its required caller-supplied registry

For a degree- d finite separable morphism, the expected mathematical identity is $\pi_*\pi^* = [d]$ on $\text{Jac}(E)$. Even after that identity is implemented, injectivity on a subgroup of prime order r requires a kernel argument; $\gcd(d, r) = 1$ is a useful sufficient route but must be checked against the exact EC1 subgroup. The delivered F2⁷/order-29 fixture now supplies that bounded check. It does not generalize the result to other catalog subgroups or create an arbitrary-class Jacobian norm.

Supported-input boundary

This audit covers only:

- Xarles arXiv:1303.4220v1;
- the two fixed PR #1383 constructor families;
- exact artifacts at the three PR heads listed above;
- the arithmetic source present at the pinned crypto baseline;
- the content-addressed feature implementation commit and local test results stated in the delivered-evidence section; and
- the EC1/ICV1 identity contracts currently documented in the repository.

Within the committed feature, generic prime Jacobian arithmetic uses deterministically validated odd primes below 2^{64} . The diagnostic linear counter is limited to $p \leq 1,000,000$, while the quadratic counter and Fp2Ctx are limited to $p \leq 4096$. Binary fields use polynomial bases of degree at most 4096. Standalone inputs are limited to 32 MiB and must match the documented JSON shapes exactly.

It does not:

- search all cover families, genera, or degrees;
- establish characteristic-three constructions;
- replace Xarles’s unknown-method comment about characteristics two and three with a nonexistence theorem;
- establish even-degree, two-infinity prime-field Jacobian arithmetic;
- certify an arbitrary-class cover-induced norm or any unlisted transfer family;
- run a Jacobian relation collection or target decomposition; or
- assess a production curve for key recovery.

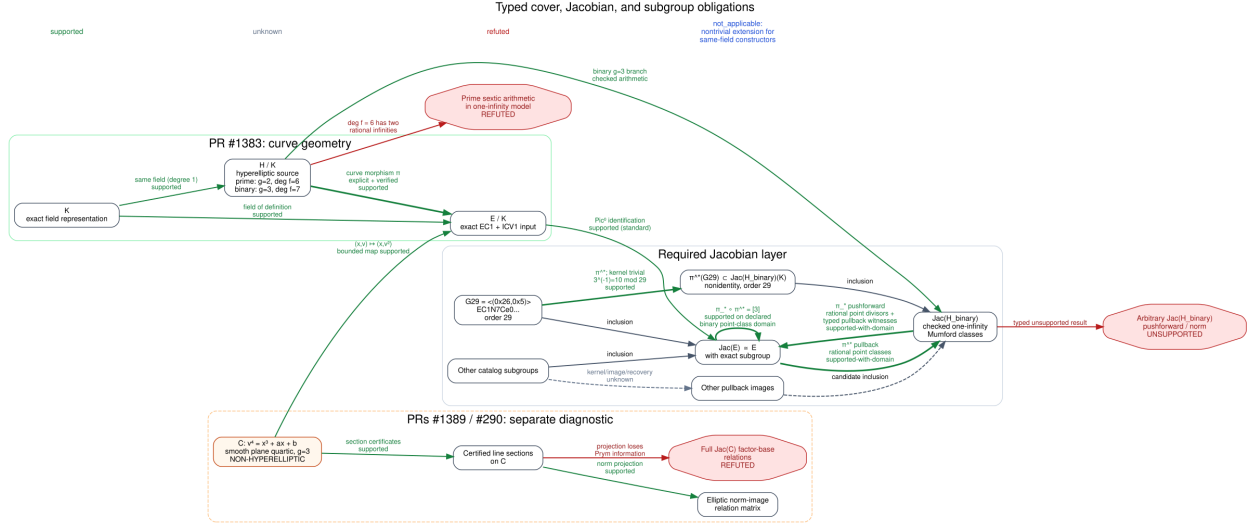


Figure 1: Typed correspondence graph

The proper negative statement is “not found or not implemented within the named family and boundary,” never “no such cover or transfer exists.”

Cost and accounting

No calibrated cost benchmark was performed for this documentation and schema round. Local correctness-test elapsed times and historical fixture counts are not converted into algorithmic costs.

Phase or quantity	Measured value	Unit	Status
Cover construction	null	null	unmeasured
Unsuccessful construction attempts	null	attempts	unmeasured
Duplicate candidates	null	candidates	unmeasured
Geometry verification	null	operations	unmeasured
Jacobian validity and reduction	null	operations	unmeasured
Jacobian addition	null	operations	unmeasured
Pullback	null	operations	bounded binary implementation present; cost unmeasured
Pushforward	null	operations	bounded binary point-divisor implementation present; arbitrary-class norm unsupported
Composition checks	null	operations	local correctness replay present; operation cost unmeasured
Subgroup/kernel checks	null	operations	exact $F_2^{2^7}$ /order-29 correctness control present; cost unmeasured

Phase or quantity	Measured value	Unit	Status
Reusable preprocessing	null	operations	unmeasured
Failed relation searches	null	attempts	not run
Relation duplicates/dependencies	null	rows	not run
Linear algebra	null	operations	not run
Target decomposition	null	operations	not run
Peak memory	null	bytes	unmeasured
Total runtime	null	milliseconds	unmeasured
Matched Pollard-rho reference	null	operations	not run
End-to-end speedup	null	ratio	not established

A missing phase makes the total and any speedup `null`, not zero.

Required controls

An implementation claiming the next status transition must include:

1. **Identity controls.** Recompute EC1 from the exact `field` and `curve` preimage; resolve the ICV1 slug and full identity against a hash-pinned registry; reject mismatches.
2. **Geometry controls.** Independently check target nonsingularity, source smoothness, genus, the rational-function identity, map degree, separability, and all places at infinity.
3. **Arithmetic controls.** Check canonicity, closure, identity, inverses, commutativity and associativity; include nontrivial partial shared support, full shared support, ramified/Weierstrass divisors, repeated support, and malformed inputs.
4. **Independent arithmetic reference.** Compare bounded exhaustive examples with a separately implemented CAS or exhaustive divisor-class oracle. A second call path through the same formulas is not independent.
5. **Transported-data control.** For exact sampled classes, compute pullback and pushforward, verify homomorphism identities, and replay $\pi_*\pi^*D = [d]D$.
6. **Coordinate-isomorphism control.** Transport a fixture through a separately specified coordinate change and check that the resulting divisor maps commute. Matching j -invariants is not enough.
7. **Subgroup control.** Bind the exact EC1 order and generator, compute or certify the kernel intersection, verify image order, and replay scalar recovery in both groups.
8. **Negative controls.** Reject the prime even-sextic in the one-infinity arithmetic path; reject composite or oversized prime moduli, invalid quadratic nonresidues, noncanonical extension elements, trailing-zero binary polynomials, altered coefficients, infinity data, maps, digests, divisors, composition multipliers, unknown model fields, and oversized command inputs.
9. **Failure accounting.** Preserve unsupported inputs, exhausted searches, duplicates, timeouts, and invalid certificates. Never turn a bounded failure into nonexistence.
10. **Cross-repository control.** Require byte-identical mirrored kernels or record a versioned adapter and both source hashes.

Full Jacobian index-calculus solver acceptance specification

This is a separate deliverable. A solver is acceptable only if all of the following are reproduced on explicitly supported parameters:

1. The source field, hyperelliptic curve, Jacobian representation, infinity configuration, subgroup order, generator, factor base, and target have exact immutable identities.
2. Every accepted relation is an equality in the **full Jacobian** and is independently replayed there. An elliptic norm image equal to zero is not a Jacobian relation certificate.

3. Relation collection records every attempt, failed decomposition, duplicate, dependent row, rejection, timeout, and stop condition.
4. The matrix modulus is the exact subgroup order. The archived matrix, column dictionary, rank target, achieved rank, dependencies, solver configuration, and solution residual are independently checked.
5. Factor-base logarithms replay against their full-Jacobian classes. Free variables, sign/orbit folding, and exceptional divisors are explicit.
6. Target decomposition is an implemented, bounded stage for a previously supplied target. It records all rerandomizations and recursive work and may not fall back to exhaustive search, rho, or a planted scalar.
7. The recovered scalar is verified by an independent multiplication $[k]P = Q$ in the claimed group. If a cover transfer is used, pullback, pushforward, composition, kernel intersection, and log recovery are separately certified.
8. Supported field/model/genus/infinity configurations and resource limits are explicit. Every other case returns structured unsupported.
9. Preprocessing, construction, unsuccessful searches, verification, duplicate handling, matrix build, linear algebra, target work, recovery, memory peak, and total runtime are accounted for in nonoverlapping intervals. Missing values stay null.
10. Any performance claim uses matched targets and resource envelopes, a verified reference solver, preserved failures, and the repository's operation-count and CI gates.

The PR #1389/#290 elliptic norm-projection experiment does not satisfy items 2, 5, 6, or 7 for the full Jacobian and must not be named as this solver.

Catalog and schema integration

The additive schema is [docs/curves/hyperelliptic-infrastructure.schema.json](#). It does not mutate the generated `covers.json` v1 artifact. It introduces:

- exact EC1 preimages and full UIDs, plus exact ICV1 registry references;
- separate characteristic-two and odd-characteristic field/model records;
- a first-class infinity configuration and Jacobian normalization context;
- a typed curve morphism distinct from induced pullback and pushforward;
- field identity/extension/restriction relationships;
- composition identities and subgroup-transfer certificates;
- exact implementation revisions and evidence digests; and
- separate claim assessments for existence, construction, verified map, arithmetic, pullback, pushforward, subgroup preservation, and computational advantage.

The conforming instance [evidence.json](#) binds the exact $F_2^{2^7}$ ICV1/EC1 target to its source, Jacobian, morphism, bounded induced-map domains, composition and order-29 subgroup certificate. It separately records the arbitrary-class norm as unsupported and the prime even-sextic one-infinity claim as refuted. Every delivered feature implementation is identified by exact commit `3b8664a466efd56121509cd77866a07e3cb55741` and source SHA-256. Historical baseline and PR implementations retain their own immutable commits.

Schema validity is not mathematical validity. A catalog exporter must also recompute hashes, resolve every local reference, verify the stated formulas, and reject dangling evidence or implementation references.

Canonical visual and dashboard review

The following canonical outputs were checked for impact:

- `docs/index-calculus-scoreboard.html`;
- `docs/ic/progress-timeline.json`;
- `docs/ic/LEADERBOARD.md` and `docs/ic/leaderboard.json`;
- `docs/browser/`; and
- `docs/curves/covers.json`.

They are unchanged. This audit adds no catalog curve, benchmark, performance value, or solver result, and the task explicitly preserves generated catalog/browser artifacts. The bounded binary point-class transfer appears only in the study-local typed graph and evidence envelope. There is no quantitative graph because every end-to-end cost is unmeasured and explicitly null.

Open obligations

1. Expand Xarles's two maps into complete rational functions on the displayed H_A , including twists and the $j = 0, 1728$ cases, then verify them.
2. Give hyperelliptic source curves and Jacobians stable content identities and emit v1 records from the native catalog exporter.
3. Extend odd-degree one-infinity validation beyond the bounded F3 fixture and compare operations with an independent CAS/divisor-class oracle.
4. Implement balanced/two-infinity arithmetic before claiming arithmetic on `prime_quadratic_pullback_v1`.
5. Compare the characteristic-two genus-three arithmetic with an independent CAS/divisor-class oracle; the exhaustive F8 control is internal.
6. Extend the bounded binary point-divisor pushforward to a verified norm on arbitrary Mumford classes, and implement balanced prime-family transfer.
7. Replay arbitrary-class composition identities after the general norm exists; the delivered infinity, exceptional, and generic point-fiber cases remain bounded evidence.
8. Produce kernel-intersection, image-order, and log-recovery certificates for every exact EC1 subgroup; the named $F2^7/\text{order-29}$ record is the sole delivered instance.
9. Pass the required CI/review gates, publish the release artifacts, and mirror accepted schemas, records, commands, and fixtures into cryptanalysis with explicit revision links.
10. Treat a full Jacobian relation solver as the separate deliverable defined above.

Reproduction notes and sources

Read-only audit commands included:

```
gh pr view 1383 --repo aburan28/crypto --json ...
gh pr view 1389 --repo aburan28/crypto --json ...
gh pr view 290 --repo aburan28/cryptanalysis --json ...
git show 7c186ca89b8efeb0db2cea53e10e6da520166b56:docs/curves/covers.json
sha256sum <exact artifacts>
jq -e . docs/curves/hyperelliptic-infrastructure.schema.json
jsonschema -i research/hyperelliptic_cover_infrastructure_20261006/evidence.json docs/curves/hyperelliptic
cargo test --lib prime_hyperelliptic::curve::tests
cargo test --lib prime_hyperelliptic::fp2::tests
cargo test --lib binary_ecc::hyperelliptic::tests
cargo test --lib binary_ecc::cover_transfer::tests
cargo test --bin curve_cover_check
cargo test --bin hyperelliptic-cover
dot -Tsvg typed_correspondence.dot -o typed_correspondence.svg
pandoc README.md --from=markdown+tex_math_single_backslash --pdf-engine=pdfelatex -o hyperelliptic_cover.pdf
pdftinfo hyperelliptic_cover_infrastructure_20261006.pdf
```

Repository sources:

- [docs/curves/HYPERELLIPTIC_INFRASTRUCTURE.md](#)
- [evidence.json](#)
- [docs/curves/COVERS.md](#)
- [docs/curves/covers.json](#)
- [docs/curves/ICV1.md](#)
- [docs/curve-identities.md](#)
- [research/bielliptic_quartic_20261005/README.md](#)

- [research/bielliptic_quartic_20261005/PROTOCOL.md](#)
- [research/bielliptic_quartic_20261005/evidence/receipt.json](#)
- [src/prime_hyperelliptic/curve.rs](#)
- [src/binary_ecc/hyperelliptic.rs](#)
- [src/cryptanalysis/jv_cover.rs](#)

No source in this round performs or plans general cryptographic key recovery.